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CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

FACULTY OF TECHNOLOGY AND ENGINEERING

DEVANG PATEL INSTITUTE OF ADVANCE TECHNOLOGY AND RESEARCH

DEPARTMENT OF INFORMATION TECHNOLOGY

A.Y 2026 – 27 Semester – 3

LAB MANUAL

CSUC201 Fundamentals of Data Structures and Algorithms

DEPSTAR

Practical-8.1

Github Link: https://github.com/rudraaaajoshi/FDSA_BATCH_A/tree/main/Practical-8

Difficulty

Understanding the four tree traversal orders was slightly challenging.

Error

Wrong traversal order can give incorrect hierarchy output, especially for level-order traversal.

Solution

Used recursion for inorder, preorder, and postorder, and a queue for level-order traversal.

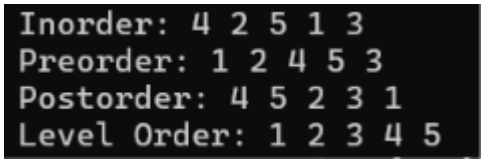
Outcome

Successfully displayed the hierarchy in all four required orders from top to bottom.

Code(Github) Link:

https://github.com/rudraaaajoshi/FDSA_BATCH_A/blob/main/Practical-8/pr-8.1.cpp

Output:



Key Question:

Problem-1:

Preorder traversal always visits the root before anything else. Using only this property — without tracing any tree — explain whether it would be possible to reconstruct the original tree from its preorder output alone, and what additional information you would need if it is not.

Answer:

No, **preorder alone is not enough** to uniquely reconstruct the original tree because it only tells us the root-before-children order. We also need the **inorder traversal** (or equivalent structural information) to determine the left and right subtrees.

Practical-8.2

Difficulty

Understanding where smaller and larger book codes should be placed was slightly challenging.

Error

If a code is placed on the wrong side, the BST property is violated.

Solution

Compare each new code with the current node: smaller → left, larger → right, until an empty position is found.

Outcome

The final inorder sequence is in ascending order, confirming that all insertions were performed correctly.

Code(Github) Link:

https://github.com/rudraaaajoshi/FDSA_BATCH_A/blob/main/Practical-8/pr-8.2.cpp

Output:

```
Book codes in sorted order: 20 30 40 50 60 70 80
Process returned 0 (0x0)   execution time : 1.045 s
Press any key to continue.
```

Key Question:

Problem-2:

The inorder traversal of a correctly built BST always produces a sorted sequence. If your inorder output is not sorted after all insertions, what does that tell you about where the error is — in the insertion logic, the traversal logic, or possibly both? How would you isolate which one?

Answer:

If the inorder output of a BST is not sorted, there may be an error in the insertion logic, traversal logic, or both. To isolate it, first check the insertion structure, then manually/independently perform inorder traversal; if the tree itself violates BST rules, insertion is wrong, otherwise the traversal is wrong.